Simple Project - 1

Objective: Capture and analyze network traffic to identify unusual patterns, malware, or signs of an attack.

Tools: Wireshark, tcpdump, Snort.

Skills Learned: Packet sniffing, traffic analysis, malware detection.

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Network Traffic Analysis using Wireshark

1. Introduction

This project focuses on capturing and analyzing network traffic using Wireshark to understand communication protocols and identify network behavior.

2. Tools Used

- Wireshark

3. Methodology

Network traffic was captured on a Wi-Fi interface while performing normal browsing activities such as accessing Google, YouTube, and Microsoft services. The captured packets were filtered and analyzed using protocol-specific filters.

4. Analysis

4.1 DNS Analysis

DNS packets were filtered using the "dns" filter. The analysis showed multiple queries for domains such as Google, Microsoft, and Bing.

Both A (IPv4) and AAAA (IPv6) records were observed, indicating support for modern networking. CNAME records were also identified, showing redirection through cloud services.

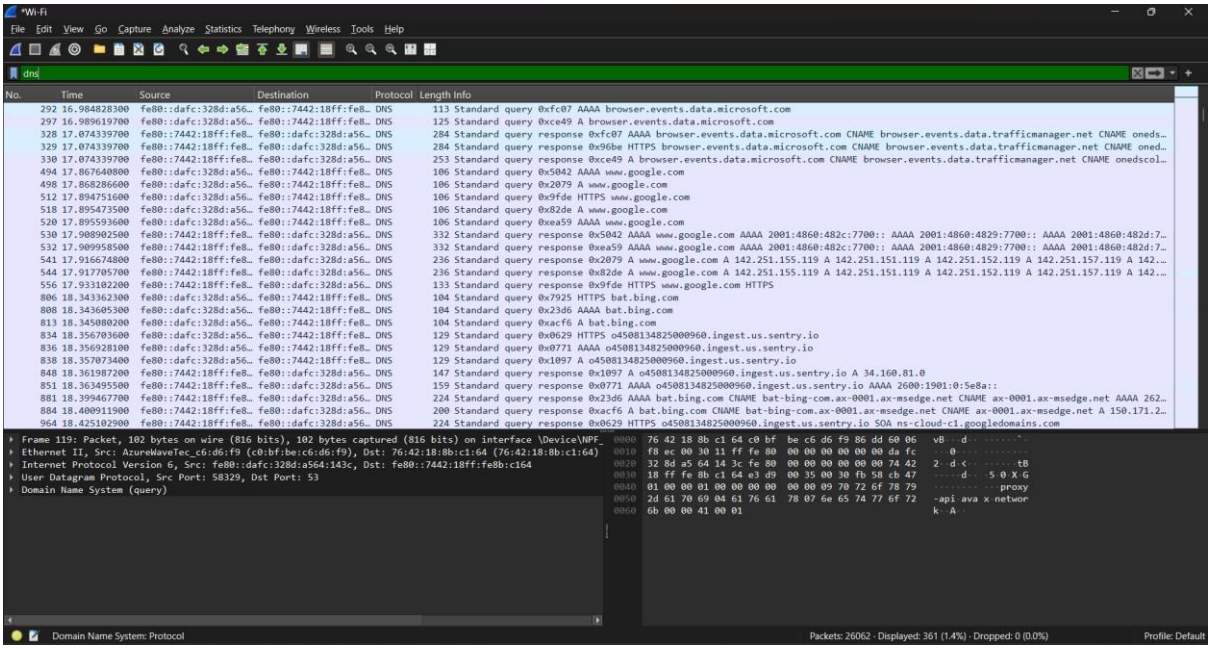

Figure 1Packet Analysis by filtering DNS

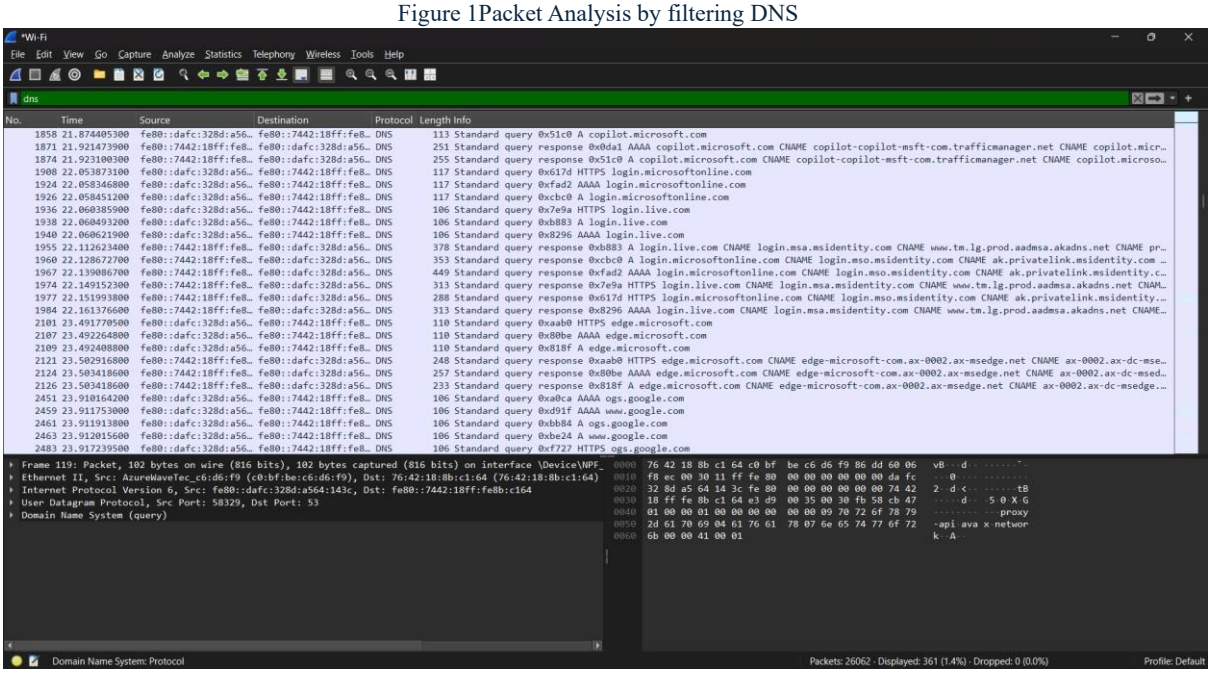

Figure 2 Packet Analysis by filtering DNS

4.2 TCP Analysis

TCP communication was analyzed using the "tcp" filter. The three-way handshake (SYN, SYN-ACK, ACK) was observed, confirming reliable connection establishment.

TCP streams showed complete communication between client and server.

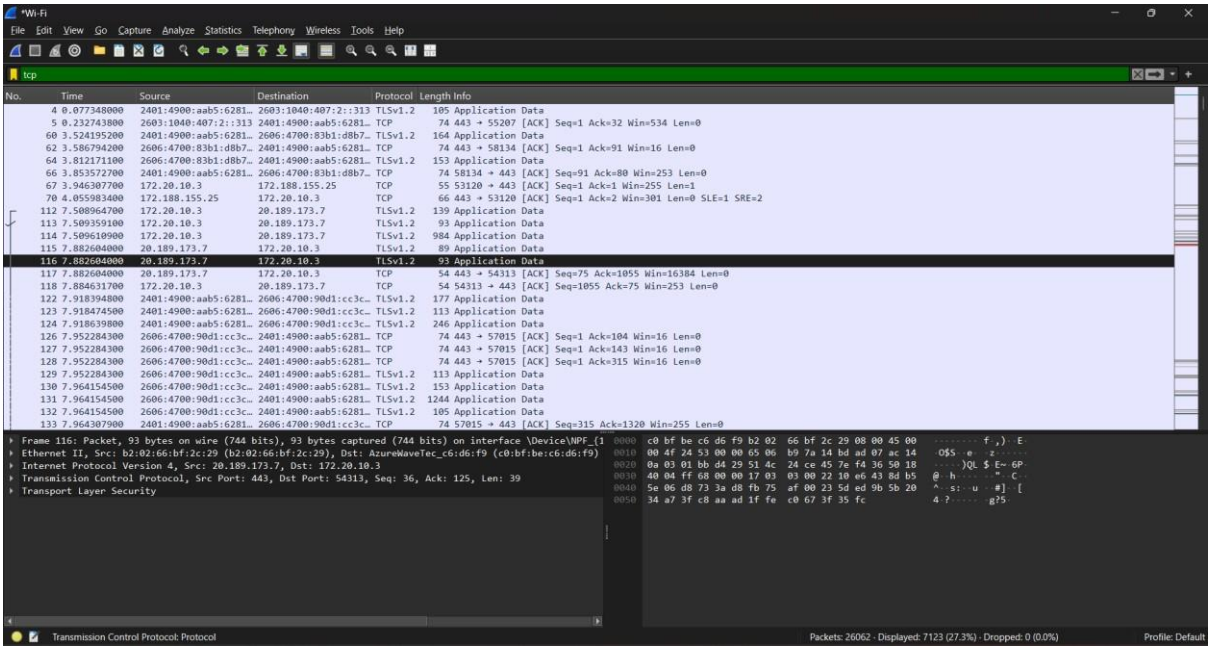

Figure 3 Packet Analysis by filtering TCP

4.3 TLS Analysis

TLS traffic was analyzed using the "tls" filter. TLS handshakes including Client Hello, Server Hello, and certificate exchange were observed.

After handshake completion, all communication appeared as encrypted application data, ensuring confidentiality.

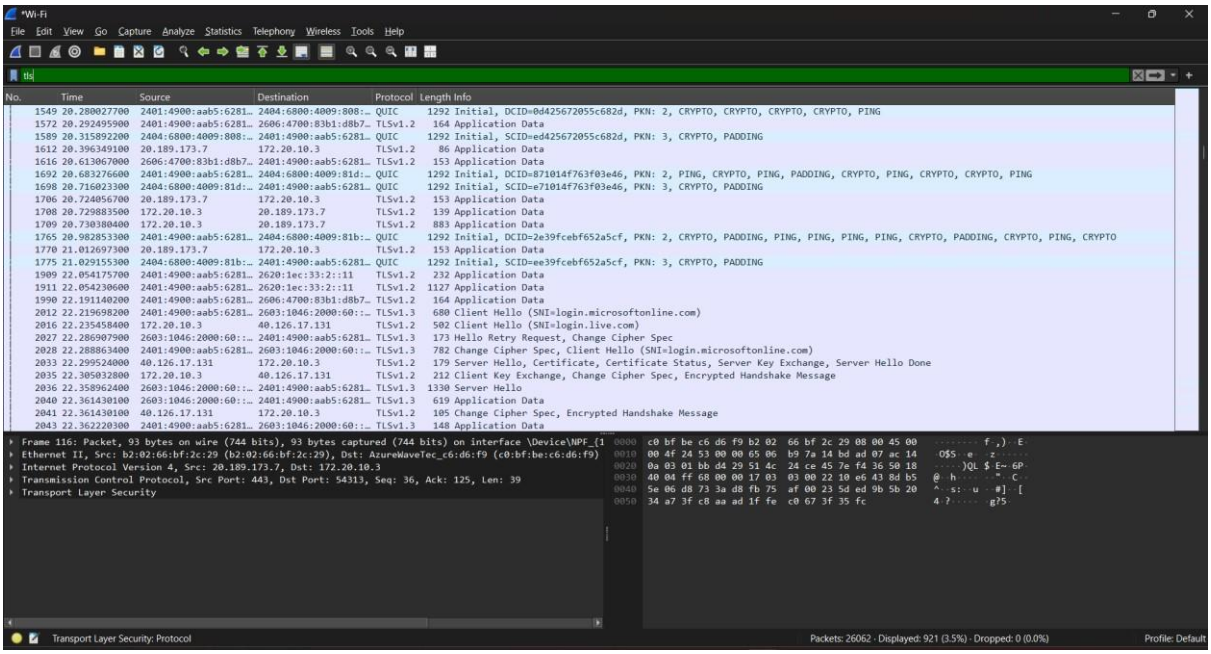

Figure 4 Packet Analysis by filtering DNS

5. Findings

- DNS resolves domain names to IP addresses
- Most traffic uses HTTPS (port 443)
- TLS ensures secure communication

- Communication with cloud/CDN servers was observed
- No suspicious or malicious activity detected

6. Conclusion

This project demonstrates how network traffic can be captured and analyzed using Wireshark. It highlights the importance of protocols such as DNS, TCP, and TLS in modern secure communication.